
title: 'DataBallPy: Load, Synchronise, and Analyse your Soccer Data tags:
- Python - football - analysis - soccer - visualisation authors: - given-name:
Gerard Alexander surname: Oonk corresponding: true orcid: 0000-0003-4056-
7274 affiliation: 1 - given-name: Daan surname: Grob affiliation: 2 - given-
name: Matthias surname: Kempe orcid: 0000-0002-4709-6172 affiliation: "1, 3"
affiliations: - name: Department of Sports Sciences, University of Groningen,
the Netherlands ror: 03cv38k47 index: 1 - name: Independent Researcher,
the Netherlands index: 2 - name: Centre for Sport Science and University
Sports, University of Vienna, Austria ror: 03prydg77 index: 3 date: 05-11-2025
bibliography: paper.bib

Summary

DataBallPy is a Python package designed to quickstart the analysis of soccer data by integrating and synchronizing event and tracking data. It provides a standardized interface for loading, preprocessing, and visualizing match data, enabling researchers and analysts to extract insights with minimal setup. By combining multiple data streams into a unified **Game** object, **DataBallPy** simplifies complex workflows and supports reproducible, data-driven soccer research. Last, **DataBallPy** is serious on documentation. Every preprocessing option and feature is elaborately explained in the documentation. **DataBallPy** does not only help in computing scientific features, but the down-to-earth documentations helps you better understand how features are computed.

Statement of need

Modern soccer analytics increasingly rely on both event data and tracking data. Event data captures specific information about events (e.g. passes and shots) like their location, success, start location, and the athlete involved in the action. This information on itself is primarily used for aggregated statistics that can be used for tactical match and player analysis (Goes et al. 2020) and is widely used in scouting because of the low cost and wide spread availability of the data (Arem et al. 2025). Tracking data, on the other hand, captures spatiotemporal information of all athletes and the ball at frequencies ranging between 10 and 25 Hz (Linke et al. 2020). This data is primarily used to quantify physical performance, but also for detection of dynamic formation (Sotudeh 2025). Current packages allow for parsing KlopPy and analysis (Raabe et al. 2022) of either datastream independently. However, there has been a growing interest in combining event and tracking data to enrich event information with spatiotemporal context. This added context provides insights and nuances, primarily on a tactical level, that neither event and tracking data can provide independently.

DataBallPy addresses this gap by combining all game related data in a stan-

standardized **Game** object. The **Game** object includes event, tracking, and metadata. The primary feature of **DataBallPy** is the robust and efficient synchronisation between event and tracking data. Although event and tracking data often both provide timestamps, their alignment has shown to be extremely poor with reported errors of 1.82 (+/-4.06) seconds. Especially the random error is concerning since it does not allow for easy correction and within 4 seconds the game might have evolved to an entirely different situation. Although specific approaches have been introduced to solve this problem, they can take between 3 and 10 minutes per game of runtime, may skip certain events, and shuffle the order of events (Roy et al. 2024; Kim et al. 2025). **DataBallPy** allows for a state of the art synchronisation algorithm that ensures the synchronisation of all events in the right order within a few seconds (Oonk et al. 2025) in just one line of code.

The synchronisation of event and tracking data allows for deeper analysis. For example **DataBallPy** provides functionality to (re)compute a frame-wise assessment of which team has ball possession, as some tracking data provides do not provide it. Furthermore, a proper analysis of goal scoring probability (xG) and expected threat (xT) is included based on the combined information of both tracking and event data which have been shown to perform better when there is a proper alignment between the event and tracking data (Oonk et al. 2025).

Next to the practical value of **DataBallPy**, as it provides low code access to scientific features, **DataBallPy** also provides as an educational tool. Often, open-source python packages provide information on how to get working code, but not on how the code works. **DataBallPy** explicitly goes a step further by elaborately explaining step by step how scientific papers are transformed into code, often referring to specific mathematical formulas as presented in the paper. This explanation is crucial since it (1) allows researchers and practitioners to better understand the strengths and weaknesses of features, and (2) teaches users on how to transform scientific papers into modular, Pythonic code. Both these characteristics provide users of **DataBallPy** a better understanding of their own analysis.

Features

The features and functionalities in **DataBallPy** can be categorised in five categories: parsing data, preprocessing, synchronisation, Performance Indicators, and visualisation.

Parsing Data

The core goal of parsing data in **DataBallPy** is obtaining a **Game** object. **DataBallPy** allows for parsing data from Tracab, Metrica, Inmotio, Opta, Instat, SciSports, Sportec, and Statsbomb internally using the `get_game` function. The **Game** object contains the event and tracking data internally as Pandas dataframes, making them intuitive to work with (team 2020).

Alternatively, one can use Kloppy to parse data from more providers and use the `get_game_from_kloppy` function to transform it to a `Game` object. Last, `DataBallPy` has included a function to load openly available data directly in a `Game` object using `get_open_game` which allows users that do not have access to data to still work with soccer data in `DataBallPy` (Bassek et al. 2025). Since parsing and the analytical pipeline of soccer data takes time and resources, `DataBallPy` can also save your processed `Game` object. Normally, raw tracking and event data together can take up to 400 MB per game, ‘DataBallPy’ downscales this to less than 20 MB per game and can be reloaded by using the `get_saved_game` function.

Preprocessing

Tracking data is often captured via video footage. Depending on the quality and number of camera’s, some noise is present in both the athlete and ball positions. `DataBallPy` allows for filtering of the tracking data, differentiation of positions to compute velocity and acceleration. Furthermore, the tracking data allows for computation of individual athlete possession (Vidal-Codina et al. 2022) and together with the event data team level possession can be estimated.

Synchronisation

`DataBallPy` uses a soccer specific implementation of the Needleman-Wunch algorithm to synchronise the event and tracking data (Oonk et al. 2025). The game can be synchronised via using the following code

```
>>> from databallpy import get_open_game
>>> game = get_open_game()
>>> game.synchronise_tracking_and_event_data()
```

Performance Indicators

`DataBallPy` has an elaborate list of scientific features included in the package. All features can be computed in a few lines of code after obtaining a `Game` object. Next to the functionality, the documentation covers an elaborate explanation of how the code works that computes the features. Using `DataBallPy` the following features can be computed:

- Covered Distance (in specific velocity and acceleration zones) (Jerome et al. 2024)
- Pressure (Andrienko et al. 2017; Herold et al. 2022)
- Individual player possession (Vidal-Codina et al. 2022)
- Expected Goals (Anzer and Bauer 2021)
- Expected Threat (Singh 2019)
- Voronoi Space Occupation (Rein et al. 2017)
- Pitch Control (Fernández et al. 2018)

Visualisation

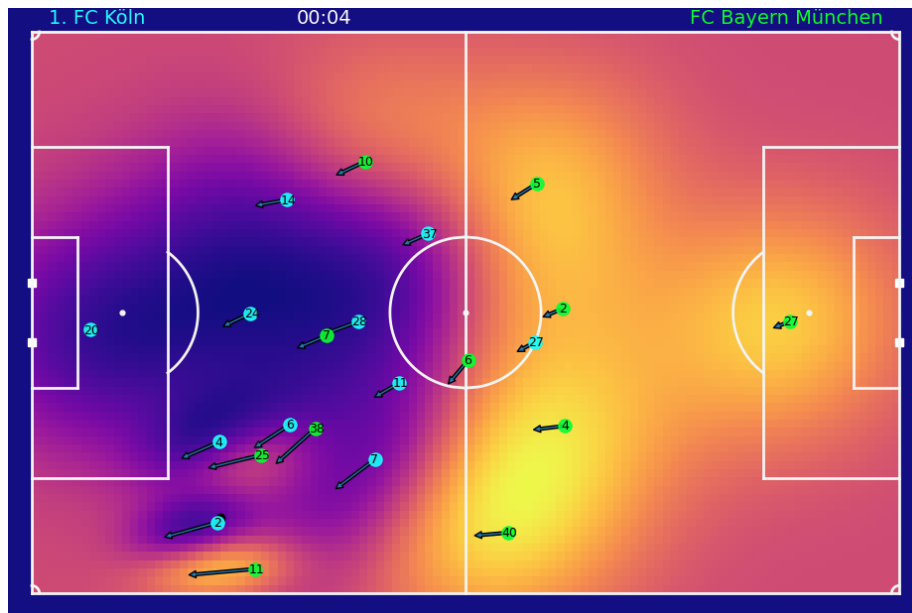


Figure 1: Example plot of soccer tracking data with pitch control heatmap as introduced in Fernández et al. (2018)

DataBallPy includes elaborate functionality to visualise the data in the `Game` object. Events locations can be visualised on a pitch using the `plot_events()` function, which allows for coloring of events by outcome, team or event type during specific periods in the game. Similarly, the locations and velocities of all players can be plotted using `plot_tracking_data()` function. If the event and tracking data is synchronised, one can also show information of the event in the same plot. Other features like pitch control heatmaps, player possession, and any custom feature can also be visualised simultaneously with the event and tracking data (Figure 1). Last, the tracking data (with heatmaps and custom features) can be transformed to a `video.mp4` to show the true spatiotemporal progression over time.

```
import matplotlib.pyplot as plt

from databallpy import get_open_game
from databallpy.visualize import plot_tracking_data

game = get_open_game()
game.tracking_data.add_velocity(game.get_column_ids() + ["ball"])

pitch_control = game.tracking_data.get_pitch_control(
```

```

        game.pitch_dimensions,
        start_idx=100,
        end_idx = 101
    )

fig, ax = plot_tracking_data(
    game,
    idx=100,
    add_velocities=True,
    heatmap_overlay=pitch_control[0],
    overlay_cmap="plasma",
    team_colors=["#00FFFF", "#00FF00"]
)
plt.show()

```

References

- Andrienko, Gennady, Natalia Andrienko, Guido Budziak, et al. 2017. “Visual Analysis of Pressure in Football.” *Data Mining and Knowledge Discovery* 31 (October): 1793–839. <https://doi.org/10.1007/s10618-017-0513-2>.
- Anzer, Gabriel, and Pascal Bauer. 2021. “A Goal Scoring Probability Model for Shots Based on Synchronized Positional and Event Data in Football (Soccer).” *Frontiers in Sports and Active Living* 3 (March): 624475. <https://doi.org/https://doi.org/10.3389/fspor.2021.624475>.
- Arem, Koen van, Floris Goes-Smit, and Jakob Söhl. 2025. “Forecasting the Future Development in Quality and Value of Professional Football Players.” *Applied Sciences* 15 (16): 8916. <https://doi.org/10.3390/app15168916>.
- Bassek, Manuel, Robert Rein, Hendrik Weber, and Daniel Memmert. 2025. “An Integrated Dataset of Spatiotemporal and Event Data in Elite Soccer.” *Scientific Data* 12 (December): 195. <https://doi.org/10.1038/S41597-025-04505-Y>.
- Fernández, Javier, F C Barcelona, Javier Fernandez, and Luke Bornn. 2018. “Wide Open Spaces: A Statistical Technique for Measuring Space Creation in Professional Soccer.” *Sloan Sports Analytics Conference*. https://www.researchgate.net/publication/324942294_Wide_Open_Spaces_A_statistical_technique_for_measuring_space_creation_in_professional_soccer.
- Goes, F R, L A Meerhoff, M J O Bueno, et al. 2020. “Unlocking the Potential of Big Data to Support Tactical Performance Analysis in Professional Soccer: A Systematic Review.” *Taylor & FrancisFR Goes, LA Meerhoff, MJO Bueno*,

- DM Rodrigues, FA Moura, MS Brink. *European Journal of Sport Science*, 2021 • Taylor & Francis 21: 481–96. <https://doi.org/10.1080/17461391.2020.1747552>.
- Herold, Mat, A Hecksteden, D Radke, et al. 2022. “Off-Ball Behavior in Association Football: A Data-Driven Model to Measure Changes in Individual Defensive Pressure.” *Journal of Sports Sciences* 40: 1412–25. <https://doi.org/10.1080/02640414.2022.2081405>.
- Jerome, Benjamin W. C., Michael Stoeckl, Ben Mackriell, Christian W. Dawson, Daniel T. P. Fong, and Jonathan P. Folland. 2024. “Contextualised Physical Metrics: The Physical Demands Vary with Phase of Play During Elite Soccer Match Play.” *European Journal of Sport Science* 24 (November): 1627–38. <https://doi.org/10.1002/EJSC.12209>.
- Kim, Hyunsung, Hoyoung Choi, Sangwoo Seo, Tom Boomstra, Jinsung Yoon, and Chanyoung Park. 2025. *ELASTIC: Event-Tracking Data Synchronization in Soccer Without Annotated Event Locations*. <https://arxiv.org/abs/2508.09238>.
- Linke, Daniel, Daniel Link, and Martin Lames. 2020. “Football-Specific Validity of TRACAB’s Optical Video Tracking Systems.” *PLOS ONE* 15: e0230179. <https://doi.org/10.1371/JOURNAL.PONE.0230179>.
- Oonk, G. A., D. Grob, and M. Kempe. 2025. “The Right Way to Synchronize Tracking and Event Data: Using Domain Knowledge to Optimize Algorithms.” In *MathSports Conference*, edited by D. Goossens.
- Raabe, Dominik, Henrik Biermann, Manuel Bassek, et al. 2022. “Floodlight - a High-Level, Data-Driven Sports Analytics Framework.” *Journal of Open Source Software* 7 (76): 4588. <https://doi.org/10.21105/joss.04588>.
- Rein, Robert, Dominik Raabe, and Daniel Memmert. 2017. “Which Pass Is Better?” Novel Approaches to Assess Passing Effectiveness in Elite Soccer.” *Human Movement Science* 55 (October): 172–81. <https://doi.org/10.1016/j.humov.2017.07.010>.
- Roy, Maaïke Van, Lorenzo Cascioli, and Jesse Davis. 2024. “ETSY: A Rule-Based Approach to Event and Tracking Data Synchronization.” *Communications in Computer and Information Science* 2035 CCIS: 11–23. https://doi.org/10.1007/978-3-031-53833-9_2.
- Singh, Karun. 2019. *Introducing Expected Threat (xT)*. <https://karun.in/blog/expected-threat.html>.
- Sotudeh, Hadi. 2025. “The Principles of Tactical Formation Identification in

Association Football (Soccer)—a Survey.” *Frontiers in Sports and Active Living* 6: 1512386.

team, The pandas development. 2020. *Pandas-Dev/Pandas: Pandas*. V. latest. Zenodo, released February. <https://doi.org/10.5281/zenodo.3509134>.

Vidal-Codina, Ferran, Nicolas Evans, Bahaeddine El Fakir, and Johsan Billingham. 2022. “Automatic Event Detection in Football Using Tracking Data.” *Sports Engineering* 25 (October). <https://doi.org/10.1007/s12283-022-00381-6>.